

Updated Report: Evaluation, Artefact Framing, and Final Artefact Validation

H&M Hybrid SVD + 3-Agent Recommendation Artefact | Updated to 20 June

1. Executive Summary

This updated report consolidates the earlier milestone report with the latest artefact, audit, validation, and packaging work completed up to 20 June. The project develops and evaluates an offline H&M recommendation artefact comparing SVD Matrix Factorisation, a structured Standalone 3-Agent recommendation workflow, and a Hybrid SVD + 3-Agent reranker.

The central academic position remains unchanged: the Standalone 3-Agent method does not replace SVD when used as an independent recommender. The agentic component is more valuable when used as part of a Hybrid reranking and explainability-oriented augmentation layer over SVD.

The newest work adds an important practical milestone. The project has moved from experiment construction into artefact delivery. The /artifact-demo route is now the main dissertation artefact page. It presents SVD, 3-Agent, and Hybrid method views using saved offline H&M experiment artefacts. Additional read-only audits confirmed that the displayed recommendation rows are not fabricated frontend records, that key metadata fields are real H&M article metadata, and that project-derived scores and explanation fields come from saved artefacts or traceable calculations.

The latest academic-risk reduction work focused on removing misleading UI wording, documenting data provenance, validating the demo, committing relevant files, and preparing the artefact ZIP for supervisor review. The project should now be presented as a controlled offline recommendation artefact with source-grounded explainability, not as a live commercial recommender or a real-time customer-engagement system.

2. Artefact Definition and Current Delivery Stage

The artefact is best defined as: H&M Hybrid SVD + 3-Agent Recommendation Artefact.

It is a runnable offline next-item recommendation demo showing SVD baseline ranking, a structured Standalone 3-Agent recommendation workflow, and a Hybrid SVD + 3-Agent reranker using real saved H&M experiment artefacts. The artefact demonstrates how user purchase history, item metadata, SVD behavioural scores, 3-Agent evidence signals, and explainability audit outputs are combined to produce and interpret Top-10 recommendations in a controlled offline evaluation setting.

The current delivery stage is artefact stabilisation and submission packaging. The core experiments have already been completed. The remaining emphasis is not to build new recommendation logic, but to validate, document, package, and submit the working artefact codebase and saved data artefacts for review.

The artefact is not a live online recommender. It does not use real-time clickstream data, live user feedback, impressions, dwell time, add-to-cart events, conversion events, or online A/B testing. For this reason, the artefact cannot be used to claim direct improvements in live customer engagement, CTR, CVR, revenue, conversion, or feedback adaptation.

3. Research Context and Motivation

The project originally explored whether an agentic recommendation approach could improve e-commerce recommendation beyond a collaborative filtering baseline. However, the formal experiment results show that a Standalone 3-Agent method is weaker than SVD when used independently. This finding required a more careful and academically honest reframing of the research.

The final research direction is: Can a structured 3-Agent recommendation workflow provide useful evidence-based signals when used as an augmentation and explainability layer on top of an SVD recommender?

This framing recognises that SVD remains a strong behavioural baseline while also identifying the value of the 3-Agent component as a source of structured preference and item evidence. The project contribution is therefore not that agentic AI is universally better than traditional recommendation systems, but that agentic evidence can support reranking and explanation when combined with a stronger collaborative filtering method.

4. Data and Evaluation Design

The project uses the H&M Personalized Fashion Recommendations dataset. The formal experiment relies on transaction records and article metadata. User demographic attributes are not used in the formal recommendation algorithms.

The formal evaluation uses a leave-one-out strategy. For each valid user, the final purchase is held out as the ground-truth next item, while the user’s previous purchases form the training history. A valid user must have at least two transactions so that both a training history and a held-out ground-truth item can be constructed.

Each evaluated user has a candidate pool of 100 items. This pool includes the held-out ground-truth item and 99 negative candidate items. The same candidate pool is shared across SVD, Standalone 3-Agent, and Hybrid. This is essential for fair comparison because differences in the results should come from the ranking logic, not from different candidate set difficulty.

The methods output Top-10 recommendations. The formal metrics are HitRate@10, NDCG@10, ILD@10, and bootstrap 95% confidence intervals. This evaluation design is suitable for an offline transaction dataset, but it cannot directly measure live customer engagement, CTR, CVR, or business conversion outcomes.

Metric	Meaning	Role in this project
HitRate@10	Whether the held-out ground-truth item appears in the Top-10 list	Measures offline next-item retrieval success
NDCG@10	Whether the held-out item appears and how high it is ranked	Measures ranking position quality
ILD@10	Intra-list diversity within the Top-10 list	Measures diversity of recommended items
Bootstrap 95% CI	Uncertainty estimate around metrics and pairwise differences	Supports cautious interpretation of method differences

5. Method 1: SVD Matrix Factorisation Baseline

SVD Matrix Factorisation is the formal collaborative filtering baseline. It uses historical user-item interaction structure to learn latent factors and rank candidate items by predicted relevance.

In this project, SVD is important because it captures behavioural collaborative filtering signals from user purchase history. This type of behavioural signal is difficult for a purely metadata-based agentic method to replace. The experimental results show that SVD is the strongest standalone offline baseline. It outperforms the Standalone 3-Agent method on HitRate@10, NDCG@10, and ILD@10 in both 1,000-user evaluations.

SVD should therefore be framed as the main behavioural benchmark. It should not be weakened or hidden in the final artefact. The artefact should make clear that the Hybrid system builds on top of SVD rather than replacing it.

6. Method 2: Standalone 3-Agent Recommendation Workflow

The Standalone 3-Agent method is a structured agentic recommendation workflow. It consists of three role-specialised components: Preference Agent, Evidence / Item Agent, and Decision Agent.

The Preference Agent extracts soft user preference signals from historical training items. These signals may include product type, product group, colour group, graphical appearance, and other structured article metadata patterns. The Evidence / Item Agent gathers structured metadata evidence for items in the shared candidate pool. The Decision Agent ranks candidate items using the preference and evidence signals, producing a Standalone 3-Agent Top-10 recommendation list without relying on the SVD score.

The current 3-Agent workflow is not a fully autonomous multi-agent system. It does not independently plan actions, call external tools, update memory from real-time user behaviour, or learn from live feedback. It is better described as a controlled, role-specialised agentic workflow designed for reproducible offline evaluation.

The Standalone 3-Agent method is valuable as an ablation. It shows what happens when the agentic evidence layer is used without SVD. The results show that this standalone approach is weaker than SVD, supporting the conclusion that the agentic component should not be positioned as a replacement for collaborative filtering.

7. Method 3: Hybrid SVD + 3-Agent Reranker

The Hybrid method is the final artefact method. It combines SVD behavioural relevance with 3-Agent evidence signals and a small diversity adjustment. The Hybrid score is defined as:

$$\text{hybrid_score} = 0.70 * \text{normalized_svd_score} + 0.25 * \text{normalized_agentic_score} + 0.05 * \text{diversity_bonus}$$

This formula makes the Hybrid method a fixed-weight heuristic reranker. It is not a learned neural hybrid model and it does not automatically learn the optimal combination of signals. The weights are manually defined and should be described honestly.

The architecture is still reasonable because it preserves the strong behavioural signal from SVD while adding agentic evidence as an interpretability and reranking signal. The 0.70 weight on SVD reflects the importance of collaborative filtering. The 0.25 agentic component allows metadata-based

preference evidence to influence ranking. The 0.05 diversity bonus adds a small adjustment, although the experimental results do not support a claim that Hybrid improves diversity over SVD.

The Hybrid method should therefore be framed as a reranking and explainability-oriented augmentation layer over SVD, not as a full replacement for SVD.

8. Explainability Audit Layer

The explainability audit layer is a supporting part of the artefact. Its purpose is to check whether Hybrid Top-10 recommendations can be explained using real source evidence. The audit uses available project data such as user purchase history, item metadata, score components, rank-shift information, preference traces, and evidence coverage.

This layer does not generate new recommendations. It does not change the ranking outputs and does not modify the evaluation metrics. It is a read-only validation and explanation layer.

The latest completed explainability audit covered the full seed99 1,000-user run, giving 10,000 recommendation explanations. The audit results show strong source-grounding under the project-defined audit rules.

Explainability audit field	Latest value / status	Interpretation
Number of explanations	10,000	Top-10 explanations across 1,000 users
Evidence coverage rate	1.0000	Explanations have available evidence fields
Preference trace rate	0.9954	Most explanations can be traced to user preference evidence
Score component coverage rate	1.0000	Score components are available
Groundedness rate	1.0000	Explanations are grounded under the audit rules
Rank-shift coverage rate	0.6974	Rank-shift evidence is available for a substantial share
Ungrounded claim count	0	No ungrounded claims were identified by the audit

These explainability metrics are project-defined source-grounding audit metrics. They are not native H&M dataset metrics and they do not measure human-perceived explanation quality, user trust, CTR, CVR, or live customer engagement. The correct claim is that the audit supports source-grounded interpretation of Hybrid recommendations within the offline artefact.

9. Formal Experiment Results

9.1 First 1,000-User Evaluation

Method	HitRate@10	NDCG@10	ILD@10
SVD	0.500000	0.295493	0.751933
Standalone 3-Agent	0.274000	0.155365	0.543800
Hybrid SVD + 3-Agent	0.498000	0.304552	0.694481

The first evaluation shows that SVD clearly outperforms Standalone 3-Agent on HitRate@10, NDCG@10, and ILD@10. This indicates that the Standalone 3-Agent method is not strong enough to replace collaborative filtering. The Hybrid method substantially improves over Standalone 3-Agent and achieves SVD-comparable HitRate@10. It also achieves a higher NDCG@10 than SVD in this

run. However, its $ILD@10$ is lower than SVD, meaning that Hybrid produces less diverse recommendation lists.

9.2 Seed99 Robustness Evaluation

Method	HitRate@10	NDCG@10	ILD@10
SVD	0.505000	0.296245	0.755089
Standalone 3-Agent	0.269000	0.154352	0.539319
Hybrid SVD + 3-Agent	0.526000	0.312443	0.692919

The second 1,000-user run using seed99 supports the same broad story. SVD again outperforms Standalone 3-Agent by a wide margin. Hybrid again substantially outperforms Standalone 3-Agent. Hybrid has a higher $HitRate@10$ than SVD in the seed99 run, but the Hybrid-minus-SVD $HitRate@10$ confidence interval crosses zero, so it is not academically safe to claim that Hybrid clearly beats SVD on $HitRate@10$.

Hybrid has a higher $NDCG@10$ than SVD in the seed99 run, and the confidence interval for the $NDCG@10$ difference remains above zero. This suggests that Hybrid may improve ranking position quality in this robustness run. However, Hybrid again has lower $ILD@10$ than SVD, and the confidence interval supports the conclusion that Hybrid is less diverse than SVD.

9.3 Cross-Run Consistency

Method	Run 1 HitRate@10	Seed99 HitRate@10	Run 1 NDCG@10	Seed99 NDCG@10	Run 1 ILD@10	Seed99 ILD@10
SVD	0.500000	0.505000	0.295493	0.296245	0.751933	0.755089
Standalone 3-Agent	0.274000	0.269000	0.155365	0.154352	0.543800	0.539319
Hybrid SVD + 3-Agent	0.498000	0.526000	0.304552	0.312443	0.694481	0.692919

Across both runs, the direction of the results is consistent. SVD consistently outperforms Standalone 3-Agent. Hybrid consistently and substantially outperforms Standalone 3-Agent. Hybrid remains competitive with SVD on ranking quality and shows stronger $NDCG@10$ in the seed99 robustness run, but it does not consistently demonstrate a statistically clear $HitRate@10$ advantage over SVD. Hybrid also consistently reduces intra-list diversity compared with SVD.

10. Overall Experimental Conclusion

1. SVD consistently outperforms the Standalone 3-Agent recommender, confirming that behavioural collaborative filtering signals remain important for H&M-style e-commerce recommendation.
2. Standalone 3-Agent is not a suitable replacement for SVD. The 3-Agent method appears to over-match narrow metadata patterns and does not generalise as effectively as SVD's behavioural signal.
3. Hybrid SVD + 3-Agent consistently and substantially outperforms Standalone 3-Agent, showing that the agentic component is more effective when used as an augmentation layer rather than as an independent recommender.
4. Hybrid remains competitive with SVD and may improve ranking position quality, especially in the seed99 run where Hybrid shows stronger $NDCG@10$ than SVD.

5. Hybrid does not consistently demonstrate a clear HitRate@10 advantage over SVD because the relevant confidence interval crosses zero.
6. Hybrid consistently reduces intra-list diversity compared with SVD, so the project must not claim diversity improvement.
7. The project cannot claim live customer engagement, CTR, CVR, conversion, or revenue improvement because the dataset and evaluation design are offline.

The final academic conclusion is that the Hybrid SVD + 3-Agent framework should be positioned as a reranking and explainability-oriented augmentation of SVD, rather than a full replacement for collaborative filtering.

11. Latest Artefact UI and Data Provenance Work

After the milestone report, the most important new work focused on making the artefact UI academically safe and source-backed. The goal was not to add new recommendation logic, but to confirm that displayed fields are supported by saved artefacts or real H&M metadata.

11.1 /artifact-demo Route

The current artefact route is /artifact-demo. It is a dark card/grid-style dissertation artefact UI with exactly three method tabs: SVD, 3-Agent, and Hybrid. The old demo reference folder is not treated as the final artefact. The /artifact-demo route is the main inspection point for the supervisor.

11.2 Source Audit of Production Demo

A read-only audit found that the production demo does not appear to fabricate recommendation rows. The demo service reads saved processed artefacts from disk, including evaluation outputs, SVD recommendation outputs, agentic recommendation outputs, Hybrid outputs, and explainability files. Synthetic records such as u1 or a4 were found only in test fixtures, not in production artefact data.

The audit also found that the demo is pinned to selected saved experiment runs. This is a hardcoded demo selector, but it is not fabricated data. It is acceptable for a reproducible dissertation artefact as long as the README explains that /artifact-demo displays selected saved formal artefacts rather than rerunning experiments live.

12. UI Risk Fixes and Field Audits

12.1 description:keyword overlap

A high-risk field audit was performed for description:keyword overlap. This phrase would be risky because it sounds like a specific text-overlap calculation based on item descriptions. The audit found no exact source field, no backend calculation, and no saved artefact value for this phrase. The current UI does not display it. Therefore, it should not be used in the artefact or dissertation unless a real source-backed calculation is implemented and verified.

12.2 Constraint Status / No hard constraints applied

The 3-Agent Final Recommendation cards previously displayed a field labelled Constraint Status with the value No hard constraints applied. This was academically risky because the project does not

implement a real hard-constraint recommendation layer. The field was removed and replaced with a real source-backed H&M metadata field: TYPE.

The corrected card layout now uses the metadata grid TYPE, GROUP, COLOUR, and APPEARANCE. This is safer because the values come from H&M article metadata fields rather than from a non-existent constraint module.

12.3 Verification of Four Metadata Tiles

UI field	Source field	Classification
TYPE	product_type_name	REAL_HM_METADATA
GROUP	product_group_name	REAL_HM_METADATA
COLOUR	colour_group_name	REAL_HM_METADATA
APPEARANCE	graphical_appearance_name	REAL_HM_METADATA

A later read-only audit confirmed that the four metadata fields are loaded through the backend payload and frontend API types, not hardcoded as fake UI values. The audit also confirmed that match score is a saved artefact value, matches count is derived from `matched_evidence.length`, score explanation text is saved from `recommendation_reason` or `explanation_text`, and evidence chips come from the saved `matched_evidence` array.

Displayed field	Verified source	Classification
MATCH SCORE	saved recommendation artifact score	REAL_SAVED_ARTIFACT
MATCHES count	<code>matched_evidence.length</code>	REAL_FRONTEND_DERIVED
Score explanation text	<code>recommendation_reason / explanation_text</code>	REAL_SAVED_ARTIFACT
Evidence chips	<code>matched_evidence</code> array	REAL_SAVED_ARTIFACT

13. README Provenance and Documentation Update

The project documentation was also updated or prepared for update so that the supervisor can understand what /artifact-demo shows. README.md was identified as the best primary location for a short artefact provenance note because it is the main repository entry point.

The provenance note explains that /artifact-demo is a runnable offline dissertation artefact demo with SVD, 3-Agent, and Hybrid method views. It states that the demo is pinned to selected saved formal experiment artefacts for reproducible demonstration, and that it does not rerun experiments or generate new live recommendations. It also distinguishes real H&M metadata fields from project-generated saved artefact outputs and presentation-layer fallback text.

This documentation is important because it protects academic honesty. It prevents the artefact from being misread as a live recommender, a live LLM generation system, or an online customer-engagement platform.

14. Validation, Commit, and Packaging Status

After the UI and documentation work, final validation and commit work were completed. The project is now in packaging mode rather than active development mode. The remaining practical task is to prepare a clean ZIP package for supervisor review.

The ZIP package should include the codebase, data-processing scripts, saved processed artefacts, saved evaluation outputs, SVD / 3-Agent / Hybrid recommendation outputs, explainability outputs, README documentation, and references or setup notes. It should exclude .git, node_modules, .venv, .next, logs, cache folders, .env files, API keys, temporary files, and unrelated reference/demo folders that are not part of the final artefact.

This packaging step is directly aligned with the latest supervisor request: the supervisor wants to inspect the codebase and data, not only watch a demo video or read the report.

15. Artefact Value

The artefact is valuable because it makes the research system visible and inspectable. It is not only a table of experiment results. It provides a working demonstration of how the methods operate on real saved H&M experiment artefacts.

The artefact should show three method views:

8. SVD tab: shows the behavioural collaborative filtering baseline and its Top-10 recommendation output.
9. 3-Agent tab: shows the structured Preference to Evidence to Decision workflow and demonstrates how user preferences are extracted, item evidence is constructed, and the standalone agentic method ranks candidates.
10. Hybrid tab: shows the final artefact method and makes the end-to-end layer relationship clear: Input / User Context, SVD Behavioural Signal, 3-Agent Evidence Signal, Hybrid Reranking Formula, Hybrid Top-10 Output, and Explainability Audit.

The artefact's value is not that it is a production recommender. Its value is that it demonstrates a controlled offline research workflow with real data, real saved recommendation artefacts, transparent method comparison, and source-grounded explanation evidence.

16. What the Project Supports and Does Not Support

16.1 Supported Claims

- SVD is a strong offline collaborative filtering baseline for this H&M experiment.
- Standalone 3-Agent is weaker than SVD as an independent recommender.
- Hybrid SVD + 3-Agent substantially improves over Standalone 3-Agent.
- Hybrid can be interpreted as a useful reranking and explainability-oriented augmentation of SVD.
- Agentic evidence is more useful when combined with SVD than when used alone.
- The explainability audit supports source-grounded interpretation of Hybrid recommendations.
- The /artifact-demo route provides an inspectable artefact backed by saved offline experiment outputs.

16.2 Unsupported Claims

- Agentic AI replaces collaborative filtering.
- Standalone 3-Agent outperforms SVD.
- Hybrid clearly and consistently beats SVD on HitRate@10.
- Hybrid improves recommendation diversity.
- Hybrid improves live customer engagement.
- Hybrid improves CTR, CVR, conversion, revenue, or add-to-cart behaviour.
- The artefact is a fully autonomous online agent system.
- The system adapts to real-time user feedback.
- The explainability audit proves real user preference or business impact.

This distinction is essential for academic integrity.

17. Objective Assessment of the 3-Agent Architecture

The 3-Agent architecture is professionally structured for an MSc-level offline recommendation artefact, but it should not be overstated as a fully autonomous agent architecture. It is professional because it has clear role separation, clear inputs and outputs, a reproducible workflow, metadata-grounded evidence, interpretable decision logic, usefulness as an ablation method, and a natural connection to the Hybrid explainability layer.

However, it is not a full autonomous agent system because it does not include autonomous planning, real-time tool use, online memory updating, live feedback adaptation, self-correction loops, multi-step agent negotiation, or production deployment. The best description remains: a structured 3-Agent recommendation workflow with role-specialised components.

18. Objective Assessment of the Hybrid Architecture

The Hybrid architecture is reasonable and aligned with the experiment results, but it should be described carefully. It combines the strong behavioural signal from SVD, interpretable metadata-based evidence from the 3-Agent workflow, a small diversity adjustment, and a clear reranking formula.

However, it is a fixed-weight heuristic reranker, not a learned adaptive model. The project did not tune the Hybrid weights or learn the optimal combination from data. The diversity bonus also does not result in higher observed diversity than SVD. The Hybrid method is therefore best framed as a controlled reranking and explainability augmentation layer over SVD, not as a new autonomous AI model that replaces collaborative filtering.

19. Limitations

11. Offline dataset only: the project uses historical transaction data and does not include real-time online behaviour.

12. No live customer engagement data: there are no impressions, clicks, dwell time, add-to-cart events, conversion events, or online A/B test results.
13. Fixed Hybrid formula: the Hybrid weights are manually defined and not learned or tuned.
14. Single dataset: the results are based on the H&M dataset and should not be treated as universal across all e-commerce domains.
15. Standalone 3-Agent underperformance: the agentic method is weak as an independent recommender and should be treated as an ablation rather than the final strongest model.
16. Diversity trade-off: Hybrid reduces intra-list diversity compared with SVD.
17. Controlled agentic workflow: the 3-Agent system is structured and role-specialised, but not fully autonomous.
18. Explainability is not user validation: source-grounded explanations support interpretability, but they do not prove that users prefer the recommendations.
19. Artefact demo is pinned to saved artefacts: this supports reproducible inspection, but it is not a live dynamic experiment runner.

20. Final Positioning

This project builds and evaluates a controlled offline H&M recommendation artefact comparing SVD, a structured Standalone 3-Agent recommender, and a Hybrid SVD + 3-Agent reranker. The results show that Standalone 3-Agent is not a replacement for SVD, but that its evidence-based signals become useful when used within a Hybrid reranking framework. The Hybrid approach should therefore be framed as an explainability-oriented augmentation of SVD, not as a full replacement for collaborative filtering.

The newest artefact work strengthens this position by making the demo source-backed, auditable, and academically honest. The risky UI fields were checked or corrected, real H&M metadata fields were verified, README provenance was added or prepared, validation and relevant commits were completed, and the project is now ready for clean ZIP packaging and supervisor inspection.